

Math 534 H - Spring 2026. Lectures 6 and 7 Part 1

①

How do we solve the inhomogeneous transport equation:

$$(ITE) \begin{cases} u_t + b u_x = F(x, t) \\ u(x, 0) = g(x) \end{cases} \quad (*)$$

where b is a scalar, $b \in \mathbb{R}$.

We know the solution to (ITE) is the sum of the solution to the homogeneous

problem (together with the auxiliary condition $(*)$ if given),

that is the solution to
$$\begin{cases} u_t + b u_x = 0 \\ u(x, 0) = g(x) \end{cases} \quad (HTE)$$

plus a particular solution to:

$$x \in \mathbb{R}, t \in \mathbb{R}$$

Similar argument works for $\mathbb{R}^d$ where one has

$$u_t + \vec{b} \cdot \vec{\nabla}_x u = F(x, t)$$

instead and $\vec{b} \in \mathbb{R}^d$

②

$$\begin{cases} u_t + b u_x = F(x, t) \\ u(x, 0) = 0 \end{cases} \quad (\text{since the auxiliary condition } * \text{ has already been taken into account when solving the (HTE)})$$

We know how to solve (HTE): the general solution is constant along the characteristic lines in the direction of the vector $\vec{V} = (b, 1)$. Let's write these lines in parametric form $\chi(s) = (x + sb, t + s)$, $s \in \mathbb{R}$. Therefore if we let $z(s) := u(x + sb, t + s) = u \circ \chi(s) \Rightarrow$

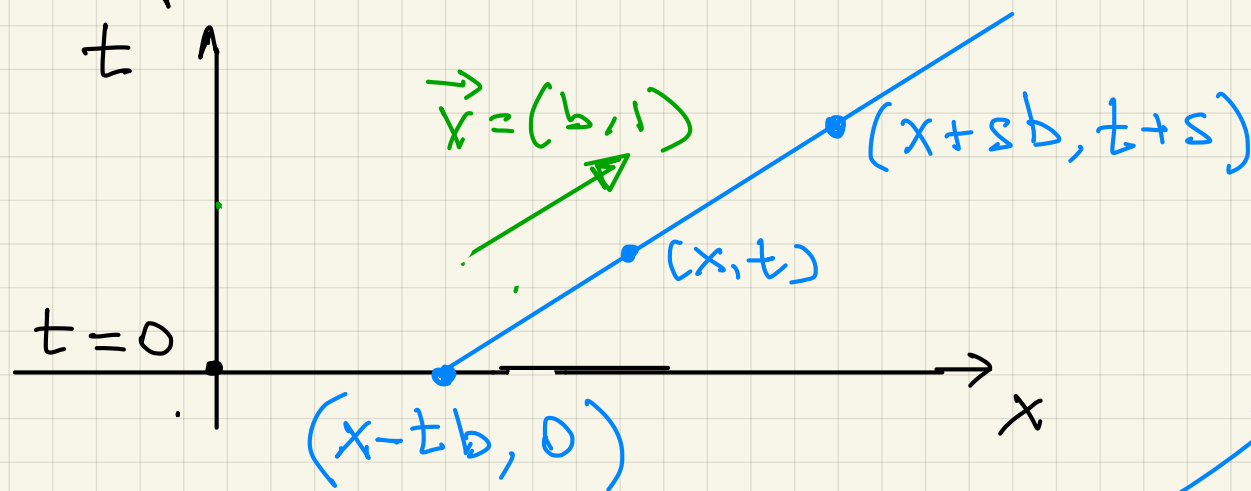
$$\frac{\partial z}{\partial s} = \frac{\partial u}{\partial t}(x + sb, t + s) \cdot 1 + \frac{\partial u}{\partial x}(x + sb, t + s) \cdot b \quad (\text{by chain rule})$$

$$= u_t(\cdot, \cdot) + b u_x(\cdot, \cdot)$$

$$= 0 \quad \text{if } \underline{u \text{ solves HTE}}, \text{ that is } \underline{\frac{\partial z}{\partial s} = 0}$$



Therefore $z(s)$ is constant on these lines (+)



The line intersects $\mathbb{R} \times \{t=0\}$ when $s = -t$ ($s+t=0$)

Choose $s=0$
 $s = -t$

From (+) then $z(0) = z(-t)$

$$u(x+0b, t+0) = u(x-tb, t-t)$$

$$u(x, t) = u(x-tb, 0)$$

$\therefore u(x, t) = g(x-tb)$ by (*); solves

the homogeneous transport equation (HTE)

Now, if we wanted to solve instead the

(ITE) we would need that $z(s) = u(x+sb, t+s)$ / ④

$$\frac{\partial z}{\partial s} = u_t(x+sb, t+s) + b u_x(x+sb, t+s). \quad \underline{\text{to}}$$

same calculation as before

$$\underline{\text{equal}} = F(x+sb, t+s) \quad (\text{instead of } = 0 \text{ as in } \textcircled{*})$$

That is we have now that $z'(s) = F(x+sb, t+s)$

So, by the Fundamental Theorem of Calculus we have
(FTC)

that

$$\underbrace{z(0) - z(-t)}_{\text{sol. homog. problem}} \stackrel{\text{FTC}}{=} \int_{-t}^0 z'(s) ds = \int_{-t}^0 F(x+sb, t+s) ds =$$

(5)

$$= \int_0^t F(x + (t-\tau)b, \tau) d\tau.$$

Change of variables $\tau := t + s$
 $d\tau = ds$

↑ is the particular solution to the inhomogeneous equation.

Hence:

DUHAMEL INTEGRAL

$$u(x, t) = g(x - tb) + \int_0^t F(x + (t-\tau)b, \tau) d\tau$$

↑ need to compute.

Solves

(ITE)

$$u_t + b u_x = F(x, t)$$

$$u(x, 0) = g(x)$$

for the given F

#

1.4 Initial and Boundary Conditions (Auxiliary Conditions) (6)

To single out one solution out of possibly infinite many we need auxiliary condition besides the PDE itself (similar as it was the case with ODE).

Two common auxiliary conditions are:

- 1) INITIAL VALUE CONDITIONS for the solution of the PDE
- 2) BOUNDARY CONDITIONS for the solution of the PDE posed on some ^(bounded) domain $\Omega \subseteq \mathbb{R}^d$.

In some cases we need to impose both 1) and 2).

- 1) Initial conditions specify the physical state of a

time dependent PDE at a particular time $t = t_0$ (7)

Examples :

a) Diffusion or Heat Equation :

$$\begin{cases} u_t = K \Delta_x u, & K > 0 \text{ constant} \\ u(x, 0) = \phi(x) \end{cases} \quad t_0 = 0 \text{ here and } \phi(x) \text{ describes the initial concentration or } \underline{\text{temperature}}$$

b) Wave Equation :

$$\begin{cases} u_{tt} = c^2 \Delta_x u & (c = \text{speed of wave, constant}) \\ u(x, 0) = \phi(x) \rightarrow \text{describes the } \underline{\text{initial position}} \\ u_t(x, 0) = \psi(x) \rightarrow \text{describes the } \underline{\text{initial velocity}} \end{cases}$$

Note that since the equation involves the second derivative in time of u we need 2 conditions to determine $u(x, t)$.

2) When PDE are defined on domains $\Omega \subseteq \mathbb{R}^d$ (8)
it is necessary to specify some Boundary conditions
for the solution to be (uniquely) determined.

In 1D Ω could be an interval; in 2D a region
in the plane, in 3D a container; etc.

The 3 most important/common boundary conditions are

a) Dirichlet BC. where one specifies u / the value
of u at the boundary $\partial\Omega$ of the domain.

b) Neumann BC. where one specifies $\frac{\partial u}{\partial \vec{n}}$ / the value
of the (external) normal derivative of u at the boundary $\partial\Omega$

of the domain

$a = \text{given constant.} \textcircled{9}$

c) Robin BC: here we specifies $\left(\frac{\partial u}{\partial \vec{n}} + a u \right) / \partial \Omega$

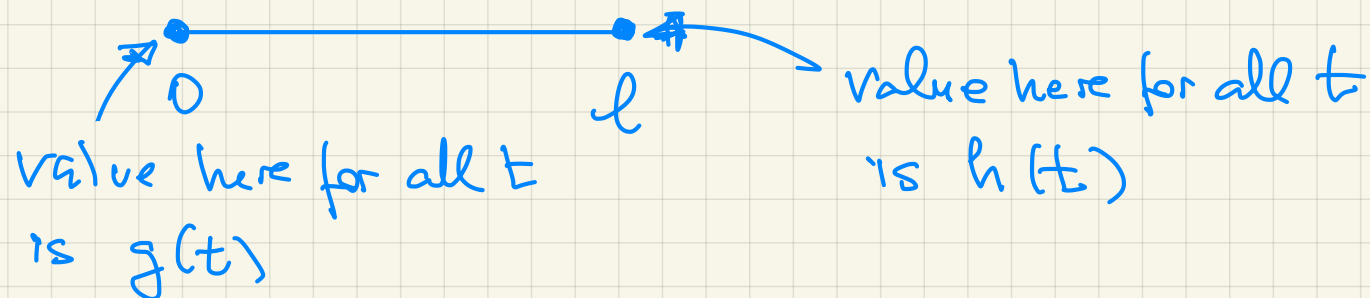
that is a linear combination of the value of the normal derivative of u and the value of u (both) at the boundary $\partial \Omega$ of the domain.

Examples

i) 1D wave eq. $u_{tt} - c^2 u_{xx} = 0, x \in [0, l]$ (string)

Dirichlet BC. would be $u(0, t) = g(t), u(l, t) = h(t)$

interval $[0, l]$:



When $g \equiv 0$ and $h \equiv 0$ we refer to these as homogeneous Dirichlet BC

On the other hand Neumann BC would be

(10)

$$\frac{\partial u}{\partial x}(0, t) = f(t); \quad \frac{\partial u}{\partial x}(l, t) = j(t)$$

normal to $[0, l]$ $\rightarrow$ differentiation in the x -direction.

In the case where the 1D wave equation describes a vibrating string then:

- If string is fixed at both ends $\Rightarrow u(0, t) = 0 = u(l, t)$
(homog. Dirichlet BC)
- If one end is free to move transversally without resistance then there is no resistance at end which gives
(say this is the end at l) $u_x(l, t) = 0$.
(homog. Neumann BC at $x = l$).
- If one end is free to move BUT it is attached to

- say - a coiled spring (obeying Hooke's law) which tends to pull back the string to the equilibrium position; then at that end we would specify Robin BC. (11)

- If one end of the string is moved in a specific way then we would have INHOMOGENEOUS Dirichlet BC at that end.

ii) Diffusion equation • if the diffusing substance is enclosed in a container Ω so that none can escape or enter then the concentration gradient in the normal direction must vanish $\otimes$ on the boundary $\partial\Omega$: that is we would have $\frac{\partial u}{\partial \vec{n}} \Big|_{\partial\Omega} = 0$ homog. Neumann BC.

($\otimes$ Fick's Law)

- if on the other hand, the container Ω is permeable.
and it is constructed so that any substance that escapes the container through the boundary $\partial\Omega$ is immediately "washed away" (think container is immersed in water)
then we must have $u|_{\partial\Omega} = 0$ (homog. Dirichlet BC)

iii) Heat equation (same eq. as Diffusion)

- recall now $u(x, t)$ = temperature at x and time t .

Let's suppose you have a rod Ω through which heat is flowing. Assume Ω is insulated along its length $x \in [0, l]$ and the end of the (hot) rod at $x = l$ is immersed in a (cool) reservoir of temperature $g(t)$. If heat

were exchanged between the end and the reservoir

(obeying Newton's cooling law) then we would have

for some $a > 0$ $u_x(l, t) + a u(l, t) = a g(t)$ Robin BC

Heat from the hot rod radiates into the cool reservoir.

Conditions at infinity on unbounded domains Ω

1) Schrödinger eq. has conservation of mass so if
we are - say - in $\mathbb{R}^3$ we'd have $\int_{\mathbb{R}^3} |u(x, t)|^2 dx = \text{constant}$

But this implies that in a $\mathbb{R}^3$
suitable sense $u(x, t)$ must "vanish" at (spatial) infinity

2) Laplace's equation $\Delta_x u = 0$ on Ω . Solutions are
minimizers of the "energy" $E(u) = \int_{\Omega} |\nabla u|^2 dx \rightarrow$ this

quantity being finite implies that in a suitable ⁽¹⁴⁾
sense ∇u "vanishes" at infinity in $\mathcal{Q} \subseteq \mathbb{R}^3$.
(there is no time here $u = u(x) \quad x \in \mathcal{Q}$)

1.5. Well-posedness of PDE

Given a PDE on a domain $\mathcal{Q} \subseteq \mathbb{R}^d$, together with a set of auxiliary conditions (initial and/or boundary conditions)

Example: vibrating string has both
on $[0, l]$

$$u(x, 0) = \phi(x)$$

$$u_t(x, 0) = \psi(x)$$

IC

$$u(0, t) = g(t)$$

$$u(l, t) = h(t)$$

BC.

We are interested mainly on 3 questions about solutions:

- Existence : There exists at least one solution a satisfying the equations + given auxiliary conditions $\left(\begin{array}{l} \text{for time dependent PDE} \\ \text{can have local in time } \mathcal{I} \\ \text{or global in time } \mathcal{I} \end{array} \right)$
- Uniqueness : There is at most one solution
- Stability The unique solution depends in a stable (continuous) manner on the data of the problem (that is, if the data are changed a little, then the corresponding solutions also change only a little").

Remarks : • If too few auxiliary conditions are imposed then there may be more than one solution (non-uniqueness) and the problem is called undetermined.

• If there are too many then there may be no solution at all (non-existence) and the problem is called overdetermined.

- (16)
- The stability property is important in physical problems because one seldom is able to measure data exactly, but up to some error. Stability guarantees that this shouldn't skew the problem.

To make precise the notion of stability one needs a norm/topology/metric in which measure "nearness"

- Non-existence ; non-uniqueness $\rightarrow$ ill-posedness

The PDE problem is then usually also unstable.

For example although the (linear) diffusion/heat eq. are generally well-posed as time dependent PDE for all $t > 0$ (FORWARD in TIME); the BACKWARDS

$t < 0$ (linear) heat equation is ill-posed (17)

Heuristically one can understand why: in diffusion or heat processes it is impossible to know what past behavior could have led up to the concentration or temperature at a later time.

We'll revisit this but here is a sketch of what happens. Let's consider the heat equation on $[-\pi, \pi)$ periodic functions. One can show that $u_t = u_{xx}$; $u(x, 0) = f(x)$ has solutions of the form

⊛ $u(x, t) = \sum_n a_n(t) \sin(nx)$ where

$$a_n(t) = e^{-n^2 t} a_n(0) \quad \text{and } a_n(0) \text{ are the coefficients}$$

(18)

of f in its sine series/Fourier series (eg. f odd periodic)

Note that when $t > 0$ the high frequency ($|n|$ large) coefficients $a_n(t) = e^{-n^2 t} a_n(0)$ decay much faster than the low frequency ones (smaller $|n|$). And in fact one can show with this information that $\sum a_n(t) \sin(nx)$ converges in a suitable sense giving rise to a (well-posed) solution.

However if $t < 0$ (backwards heat eq.) we instead have that $a_n(t) = e^{n^2 \tilde{t}} a_n(0)$ where $\tilde{t} = -t > 0$
 $\rightarrow \infty$ as $n \rightarrow \infty$

so higher frequencies blow up as $n \rightarrow \infty \forall t < 0$